PI2399 network from Small Timer SMD for ALC (all basic parts except the PI2399)
PI2399 delay, effects loop in the feedback (SEND / switched RETURN). TONE on every repeat. TIME m

**Super-Freq**

File: loopEcho.kicad_sch
Sheet: /

**Title: Loop Echo**

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| Size: A1 | Date: 2026-10-25 |
| KiCad E.D.A. 10.0.4 | Rev: 1.0 |
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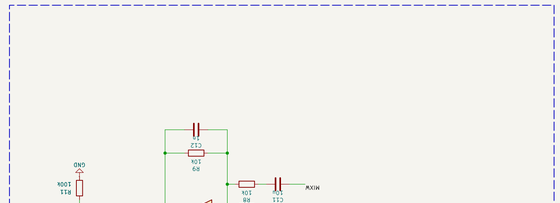
LOOP footswitch on: RET = SEND (internal path). SEND jack grounded. On: RET = RETURN jack: no plug in RETURN = its switch ties it to SEND. TONE acts on every repeat in all cases.

A fuzz in the loop adds gain every pass: the 1N5148W pair caps the loop at about +-0.6 V.

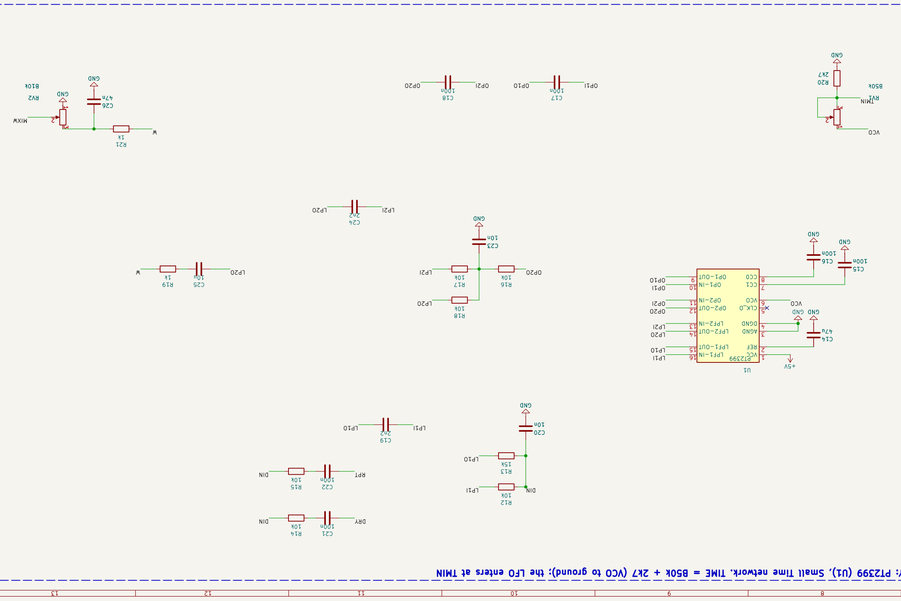
## POWER (SS14, AMS1117-5.0 for the PI2399), VB, STATUS LED, FOOTSWITCH HEADER, JACKS




## INPUT BUFFER (U2A) + OUTPUT MIXER (U2B); as Small Time




## DELAY: PI2399 (U1), Small Time network. TIME = 850k + 2k7 (VCO to ground); the LFO enters at TMIN

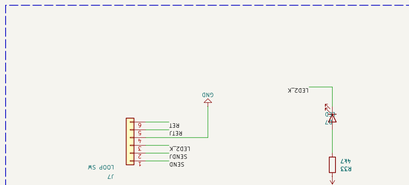



## FEEDBACK LOOP: SEND (U2A) -> loop -> RETURN (no plug = internal) -> TONE -> U3B -> limiter -> REPEAT




## LOOP FOOTSWITCH (2nd SwitchPCB v2): RET = SEND or RETURN

SwitchPCB jack: O/ = OUT when engaged; I/ when bypassed; IN = I/ when engaged; GND when bypassed. LED to GND when engaged.




## MODULATION: LFO (U4, LM358) -> DEPTH -> 33k + 10u -> TMIN (even depth across TIME). ~0.5-12 Hz

